Action-Conditioned Predictive Consistency as a Diagnostic for Gaussian-Noise Robustness in JEPA World Models

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Abstract

Latent predictive world models avoid pixel reconstruction, but pixel-free prediction is not a robustness definition for closed-loop control. We study this gap with action-conditioned predictive consistency (ACPC), a no-retraining diagnostic for JEPA-style control checkpoints. The diagnostic combines ACPC Tail Risk (ATR), which measures high-quantile clean/noisy rollout disagreement for the same state under the same action sequence, with Selective Margin Pass Rate (SMPR), which evaluates whether task-grounded near-boundary distinctions remain separated. Across three independent LeWM training seeds and four control tasks, no-noise checkpoints are fragile under observation-only Gaussian noise, while input-side noise training yields broad matched-Gaussian recovery bands under the specified stressor. Across the full training-noise sweep, recovered rows occupy low-ATR/high-SMPR regions, and held-out seed/task validations keep recovery-onset errors within two grid steps. These results support ATR and SMPR as bounded ACPC diagnostics for matched-Gaussian recovery bands in fixed JEPA world-model checkpoints.

Keywords: visual world models; JEPA; action-conditioned predictive consistency; visual robustness; predictive-dynamics diagnostics.

Code and data release: `https://github.com/Anguo-star/lewm-acpc-diagnostics`

1 Introduction

JEPAs predict future representations instead of reconstructing pixels, which can reduce pressure to encode high-magnitude irrelevant visual detail [1, 3, 29]. For control, however, "irrelevant" is action- and transition-dependent. A representation may ignore a background texture but must preserve a contact pose, doorway state, or object-goal relation if that distinction changes the next useful action. Encoder-level invariance is therefore an incomplete robustness target: planning uses the model's *predictions*, so the robustness question should be asked after action-conditioned rollout.

We study this gap as a controlled diagnostic problem for JEPA latent world-model control. Let a clean history and a visually perturbed history describe the same underlying state, and evaluate both branches under the same action sequence. The desired behavior is not necessarily identical encoder latents. It is that task-relevant predicted futures agree after the shared action intervention, while states that differ in action, transition, cost, or goal relation remain separable. This selective requirement is the core of action-conditioned predictive consistency (ACPC).

The selective tension matters because consistency alone can be misleading. A collapsed model can make clean and corrupted views agree while erasing distinctions needed for planning. Conversely, a large encoder shift can be harmless if the action-conditioned predictor contracts nuisance directions before they affect the predicted rollout used by planning. The diagnostic must therefore pair a same-state predictive-stability metric with an anti-collapse margin criterion for task-grounded near-boundary separations.

We instantiate this diagnostic with a fixed Gaussian intervention. Across PushT, TwoRoom, Reacher, and Cube, we evaluate no-noise LeWM [19] checkpoints and full-sequence input-side Gaussian-noise checkpoints over $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$. The primary endpoint is observation-only Gaussian evaluation noise with the goal image kept clean. Across three independent LeWM training seeds, the no-noise baseline remains strong on clean observations but loses 74.56 ± 5.55 points on PushT and 41.00 ± 1.44 points on Reacher at evaluation noise

$\sigma = 0.08$. Test-time visual fragility is not new [15, 33, 12, 21]; our focus is what it reveals about fixed JEPA world-model checkpoints.

The evidence shows broad matched-Gaussian recovery bands rather than a stable universal point optimum. Input-side noise training recovers the observation-noise endpoint on TwoRoom, PushT, and Reacher and gives a weaker but positive Cube signal across training seeds. At recovered endpoints, ACPC Tail Risk (ATR) is substantially lower and Selective Margin Pass Rate (SMPR) is near one across tasks. Bounded blur/resize evaluations delimit perturbation scope.

This paper makes three contributions. First, it formulates visual robustness for JEPA world-model control as selective action-conditioned predictive consistency rather than encoder invariance. Second, it derives an ACPC radius–margin diagnostic certificate and a local Gaussian sensitivity interpretation: high-probability same-state predictive radius, planner candidate margins, and task-grounded discriminability margins jointly bound fixed-pool candidate instability and collapse risk, while the composed encoder–predictor Jacobian explains how the radius can contract under matched Gaussian training. ATR and SMPR are empirical estimates of the two sides of this certificate. Third, it provides a fixed-checkpoint Gaussian robustness study across four tasks and three LeWM training seeds, including full-sweep diagnostic trajectories, held-out diagnostic-region validation, sample-level fixed-pool event-rate recomputation, finite-difference and exact-JVP/Hutchinson Gaussian sensitivity analyses, joint ATR-plus-guard validation, and bounded non-Gaussian stressor evaluations that delimit perturbation scope.

2 Related Work

This diagnostic sits between latent-prediction representation learning and robust visual-control methods. The related work is organized around three questions: what latent prediction promises, where augmentation-based invariance is insufficient for control, and how control-relevant representation work treats downstream predictive consequences.

2.1 Latent prediction and JEPA world models

JEPA [16] predicts in latent space rather than reconstructing pixels; I-JEPA [1] and the V-JEPA family [3, 2, 20] extend this idea to images, video, and dense physical-world representations. LeWM [19], our baseline, stabilizes an end-to-end JEPA world model with SIGReg [6]. PLDM [25, 26], via `stable-worldmodel` [18], provides related latent-dynamics planning context. LeJEPA theory [14] studies latent-variable recovery and latent planning; our diagnostic asks whether learned visual world-model checkpoints give paired clean/noised views consistent action-conditioned predictions.

2.2 Invariance, augmentation, and visual robustness

Augmentation and joint-embedding losses impose invariances only when their training signal identifies the right nuisance factors [31, 27, 35]. Recent theory clarifies why latent prediction can help with nuisance or noisy features but also why it is not enough for control. Van Assel et al. [29] and Littwin et al. [17] analyze representation-level effects; they do not address closed-loop action-conditioned rollout, candidate-cost stability, or the need to keep action-distinct states separable. Seq-JEPA [8] addresses a related invariance–equivariance tension architecturally.

We do not train robust visual-control systems as competing baselines. DrQ/DrQ-v2 [15, 33], SODA/DMC-GB [12], DreamerV3 [10], TD-MPC2 [11], and ViGMO [21] define future method-comparison axes; noisy/video JEPA variants [22, 13, 23] are representation-level context. ACPC is complementary: it tests whether same-state visual perturbations become equivalent after a fixed action intervention while action-, transition-, or cost-distinct cases remain discriminable.

2.3 Control-relevant predictive representations

Bisimulation merges states with equivalent reward and transition behavior. DeepMDP [7], DBC [34], bisimulation-regularized MPC [24], and Bisim-JEPA [28] use this principle to learn control-relevant representations. Value-equivalent and value-aware models [9, 30] and VIBR [5] likewise emphasize downstream control consequences. Wang et al. [32] formalize action-conditioned video world modeling as a group action with identity, inverse, and composition-consistency metrics. ReOI [4] treats visual distractor robustness in visual MPC through observation interventions for world-model action-outcome prediction. These lines show that downstream predictive consequences are already central to control-representation work. Our diagnostic instantiates that view for paired clean/noised JEPA latent-world-model rollouts: same-state noised and unperturbed views should agree after the same action sequence, while action-distinct transitions remain separable.

Our use of ACPC is therefore narrower than an identifiability theorem, a learned bisimulation metric, or a robust MPC intervention. LeJEPA-style results motivate latent prediction and planning under latent recovery assumptions; bisimulation and value-aware models define task abstractions; group-action metrics test action faithfulness; ReOI modifies observation handling for robust visual MPC. This paper instead supplies a JEPA checkpoint diagnostic centered on additive Gaussian pixel-noise stress tests: predictive stability is tested on paired same-state clean/noised rollouts under the same action sequence, fixed-candidate cost stability is derived as a sufficient condition, and a separate discriminability guard evaluates whether action-, transition-, or cost-distinct cases have been preserved.

3 Action-Conditioned Predictive Consistency

This section defines action-conditioned predictive consistency and its discriminability guard. We use ACPC to separate two requirements that encoder invariance alone conflates: same-state visual perturbations should agree after action-conditioned rollout, while task-grounded different-state pairs should remain separable. Section 3.3 then maps these two requirements to ATR and SMPR.

3.1 Setup and definitions

Let o_t be an unperturbed observation and $\tilde{o}_t = \tau(o_t)$ a visually perturbed view of the *same* underlying state (τ is additive Gaussian pixel noise here, but the definitions are perturbation-agnostic). Let h_t and $\tilde{h}_t$ be the corresponding observation–action histories, E_θ the encoder, and F_θ the action-conditioned latent predictor. Fix an action sequence $\mathbf{a}_t^H = a_{t:t+H-1}$ that is *identical* on both branches. For compact notation below, $\mathbf{a}_{0:k-1}$ denotes the length-k prefix of this sequence after re-indexing time at t, and we write

$$z_t = E_\theta(h_t), \quad \tilde{z}_t = E_\theta(\tilde{h}_t), \qquad \hat{z}_{t+k} = F_\theta^k(z_t, \mathbf{a}_{0:k-1}), \quad \hat{\tilde{z}}_{t+k} = F_\theta^k(\tilde{z}_t, \mathbf{a}_{0:k-1}). \tag{1}$$

Let Π denote a *task-relevant predictive projection* — the full latent, a transition delta, goal-distance or cost features, or a learned projection — and let d be a distance on its range.

Action-conditioned predictive consistency. For a same-state perturbation pair under the shared action sequence,

$$\mathrm{ACPC}_H(o_t, \tilde{o}_t, \mathbf{a}) = \sum_{k=1}^{H} \alpha_k\, d\big(\Pi(\hat{z}_{t+k}),\, \Pi(\hat{\tilde{z}}_{t+k})\big), \qquad \alpha_k \geq 0. \tag{2}$$

For the theoretical statements below, we use the induced horizon metric

$$d_H\big((u_1, \ldots, u_H), (v_1, \ldots, v_H)\big) = \sum_{k=1}^{H} \alpha_k d(u_k, v_k), \tag{3}$$

so that Equation (2) is exactly the clean/perturbed projected-rollout distance under the shared action sequence. In this diagnostic framing, robustness corresponds to small ACPC_H together with action-relevant discriminability, not to the encoder distance $d(z_t, \tilde{z}_t)$ alone. Encoder geometry remains an indispensable first-stage risk signal: noisy views should stay in the same-state predictive basin and avoid crossing into task-distinct neighborhoods. The narrower point is that raw encoder distance alone is not a complete robustness criterion, because the effect of an encoder shift depends on local action-conditioned rollout sensitivity and candidate margins.

Action-relevant discriminability. Consistency must be *selective*. Let $Y^H(s, \mathbf{a})$ denote a task-relevant future signal that is external to the learned predictor — simulator state, cost-to-go, inverse-dynamics label, contact/keyframe tag, or another diagnostic proxy available in the dataset. For two genuinely different states s_i, s_j (with latents z_i, z_j) that admit an action sequence under which this external future signal diverges,

$$\Delta_{\mathrm{dyn}}^H(s_i, s_j, \mathbf{a}) = d_Y\big(Y^H(s_i, \mathbf{a}),\, Y^H(s_j, \mathbf{a})\big) > m, \tag{4}$$

the representation and predictions should keep them separable,

$$d\big(\Psi(z_i),\ \Psi(z_j)\big) > m', \tag{5}$$

where Ψ is a discriminability projection, possibly action-conditioned (latent, predicted delta under $\mathbf{a}$, inverse-dynamics feature, cost feature, or rollout embedding), and $m, m' > 0$ are margins. Equations (2)–(5) state the target: lower predictive disagreement for same-state visual perturbations must preserve discriminability among state/action/transition-distinct pairs. A method that drove $\mathrm{ACPC}_H \to 0$ by collapsing the latent would violate (5), so the two conditions are inseparable.

The same guard can be posed over state–action pairs: if two pairs have external task-relevant futures that differ by more than a margin, their projected rollouts should remain separated. This includes the same clean state evaluated under two dynamically distinct actions. The reported SMPR instantiates the guard with task-grounded near-boundary state pairs rather than adding a separate action-faithfulness metric.

3.2 Predictive tubes and task margins

The selective condition can be written as a radius–margin event. For a fixed action sequence, define the projected rollout map

$$G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a})) \tag{6}$$

and the same-state predictive radius under perturbation level σ as

$$R_\sigma(h, \mathbf{a}) = d_H\big(G_{\mathbf{a}}(E_\theta(h)), G_{\mathbf{a}}(E_\theta(\tilde{h}_\sigma))\big)\,, \qquad r_{1-\alpha}(\sigma) = Q_{1-\alpha}\big(R_\sigma(h, \mathbf{a})\big). \tag{7}$$

The radius side asks whether a visual perturbation remains inside the same-state predictive tube after the action intervention. The margin side asks whether planner and task-grounded distinctions remain outside that tube. For a fixed candidate pool $\mathcal{A}$, the clean planner margin is

$$\Delta_{\mathcal{A}}(h, g) = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g), \tag{8}$$

where lower cost is preferred and $\mathbf{a}^{(1)}, \mathbf{a}^{(2)}$ are the clean top-1 and top-2 candidates. For task-grounded different-state pairs, we write the discriminability margin as

$$M_{\mathrm{diff}}(i, j, \mathbf{a}) = d(\Psi(z_i), \Psi(z_j))\,, \tag{9}$$

and track the failure event $M_{\mathrm{diff}} \leq R_\sigma + \delta_m$ for a reporting margin $\delta_m \geq 0$. Robustness requires both sides: small radius without margin is collapse, while large margin without small radius is visually unstable planning.

3.3 Operational consistency diagnostics

The reported diagnostic uses two metrics matched to this radius–margin logic. ACPC Tail Risk (ATR) estimates the upper quantile of the same-state predictive radius in Equation (7): it summarizes normalized clean/noisy ACPC-H rollout disagreement under a fixed recorded action sequence; lower is better. We use the 90th percentile in all experiments as a fixed empirical reporting choice, not a theoretical constant. Selective Margin Pass Rate (SMPR) estimates the margin-preservation side: it is the pass rate of task-grounded label-crossing pairs whose clean different-state rollout distance exceeds the same-state noisy radius at margin zero; higher is better.

The diagnostic gate is therefore not ATR alone. ATR tracks the high-tail term $\Pr[D > \epsilon]$ in the sampled-pool tail argument, while SMPR estimates whether task-grounded near-boundary pairs remain outside the perturbation tube. Low ATR without high SMPR is not interpreted as robustness, because a collapsed encoder–predictor can also make clean and corrupted views agree. Encoder geometry and planner-side analyses remain useful supplementary diagnostics, but the diagnostic used in this paper is the joint radius–margin event.

3.4 ACPC as a fixed-candidate stability condition

ACPC is connected to planning through the candidate costs evaluated by model-predictive control. Let h be a clean history, $\tilde{h}$ the corresponding perturbed history, and let both branches evaluate the same candidate action sequence $\mathbf{a}$. For a goal or task descriptor g, define

$$C_h(\mathbf{a}, g) = J(\Pi(\hat{z}_{t+1}), \ldots, \Pi(\hat{z}_{t+H}), g)\,, \tag{10}$$

where J is the planner cost on the projected-rollout sequence. $C_{\tilde{h}}(\mathbf{a}, g)$ is defined analogously from the perturbed branch. The following statements are sufficient conditions for stability on a fixed candidate set; they do not prove stability of the full CEM sampling distribution or of the closed-loop trajectory over repeated replanning.

Encoder geometry enters the planning link through the composed rollout map

$$G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a})).$$

Smaller same-state encoder shifts and reduced neighborhood crossing lower upstream risk, but the planning-relevant quantity is the projected-rollout distance after the shift is passed through $G_{\mathbf{a}}$. The empirical diagnostic therefore reports action-conditioned rollout-disagreement tails rather than raw encoder distances alone.

If the planner cost J is locally L_J-Lipschitz on a fixed clean/perturbed projected-rollout neighborhood, then a paired rollout discrepancy at most ϵ implies candidate-cost drift at most $L_J\epsilon$:

$$d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))\big) \le \epsilon \quad \Rightarrow \quad |C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \le L_J\epsilon.$$

For a shared candidate set, if every candidate cost moves by at most η and the clean top-1/top-2 margin Δ is larger than 2η, the top candidate is unchanged. Equivalently, the ACPC-cost condition gives fixed-candidate stability when $\Delta > 2L_J\epsilon$. These are fixed-candidate statements only; adaptive sampling, repeated replanning, and environment feedback can still change the closed-loop trajectory. Formal statements and proofs are in Appendix A.

Theorem 1 (ACPC radius–margin certificate)**.** *Let $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\} \sim q^K$ be a once-sampled candidate pool with deterministic tie-breaking. For a clean history h and matched perturbed history $\tilde{h}$, define*

$$D_j = d_H\big(G_{\mathbf{a}^j}(E_\theta(h)), G_{\mathbf{a}^j}(E_\theta(\tilde{h}))\big).$$

Assume the planner cost J is locally L_J-Lipschitz on the evaluated projected-rollout neighborhood and that for a single sampled candidate

$$\Pr_{\mathbf{a}\sim q}[D(\mathbf{a}) > \epsilon] \le \alpha.$$

Let $\Delta_{\mathcal{A}} = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g)$ be the clean top-1/top-2 candidate margin. Then

$$\Pr_{\mathcal{A}}\left[\arg\min_j C_h(\mathbf{a}^j, g) \ne \arg\min_j C_{\tilde{h}}(\mathbf{a}^j, g)\right] \le K\alpha + \Pr_{\mathcal{A}}[\Delta_{\mathcal{A}} \le 2L_J\epsilon].$$

If the task-grounded discriminability failure event satisfies

$$\Pr[M_{\mathrm{diff}} \le R_\sigma + \delta_m] \le \gamma,$$

then the selective-ACPC diagnostic failure risk is controlled by

$$\Pr[\text{selective failure}] \le K\alpha + \Pr_{\mathcal{A}}[\Delta_{\mathcal{A}} \le 2L_J\epsilon] + \gamma.$$

This is a diagnostic certificate for a fixed candidate pool and a matched perturbation distribution. It says that a checkpoint is in the safe radius–margin regime only when the high-probability same-state predictive radius is below the relevant planner margin and task-grounded separations remain outside the perturbation tube.

Definition 1 (Matched-perturbation diagnostic region)**.** *For a checkpoint family indexed by training perturbation level ρ and evaluated at observation noise σ, define the empirical radius–margin pass event*

$$\widehat{\mathcal{G}}_{\rho,\sigma} = \left\{\widehat{Q}_{1-\alpha}(R_\sigma) \le \widehat{r}_{\mathrm{margin}}\right\} \cap \left\{\widehat{\mathrm{SMPR}}_{\rho,\sigma} \ge 1 - \widehat{\gamma}\right\}.$$

The matched-perturbation diagnostic region is the set of checkpoint indices ρ for which $\widehat{\mathcal{G}}_{\rho,\sigma}$ holds. If planner-margin traces are unavailable, $\widehat{r}_{\mathrm{margin}}$ must be reported as a task-grounded rollout-space proxy, not as a planner-margin certificate. If diagnostics are available only at the no-noise and recovered endpoints, the evidence is endpoint mechanism evidence rather than interval identification.

3.5 Sampled-pool tail motivation and Gaussian sensitivity

The fixed-candidate statements motivate the deterministic link from paired rollout agreement to candidate decisions. A once-sampled candidate pool adds a tail requirement: if a single candidate has $\Pr[D > \epsilon] \le \delta$, then a pool of size K can still contain a bad clean/noisy rollout pair with probability up to $K\delta$. A flip can also occur when the clean top-1/top-2 margin is at most $2L_J\epsilon$. This is the reason ATR uses an upper-tail summary rather than a mean disagreement score, and it is also why ATR is not a closed-loop control guarantee. Appendix A gives the proof.

The Gaussian stressor also has a local sensitivity interpretation. Let $G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a}))$ and let $\tilde{o} = o + \xi$ with $\xi \sim \mathcal{N}(0, \sigma^2 I)$.

Proposition 1 (Local Gaussian ACPC sensitivity)**.** *If E_θ and $G_{\mathbf{a}}$ are locally twice differentiable around o and $E_\theta(o)$ with bounded second-order remainders, then for small σ,*

$$G_{\mathbf{a}}(E_\theta(o+\xi)) - G_{\mathbf{a}}(E_\theta(o)) = J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)\xi + R_\xi,$$

where the remainder contributes higher-order terms, and

$$\mathbb{E}_\xi\left[\|G_{\mathbf{a}}(E_\theta(o+\xi)) - G_{\mathbf{a}}(E_\theta(o))\|_2^2\right] = \sigma^2 \left\|J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)\right\|_F^2 + O(\sigma^3).$$

The same linearization gives the local tail interpretation used by ATR.

Corollary 1 (Local Gaussian ACPC radius quantile)**.** *Under the local linearization in Proposition 1, let*

$$A_{h,\mathbf{a}} = J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o),$$

and let $\lambda_1, \ldots, \lambda_r$ be the nonzero eigenvalues of $A_{h,\mathbf{a}}A_{h,\mathbf{a}}^\top$. For isotropic pixel noise $\xi \sim \mathcal{N}(0, \sigma^2 I)$,

$$R_\sigma^2(h, \mathbf{a}) \approx \sigma^2 \sum_{i=1}^{r} \lambda_i \chi_i^2,$$

and therefore

$$r_{1-\alpha}(\sigma; h, \mathbf{a}) = Q_{1-\alpha}(R_\sigma) \approx \sigma \sqrt{Q_{1-\alpha}\left(\sum_i \lambda_i \chi_i^2\right)}.$$

This corollary connects the empirical ATR quantile to the singular spectrum of the action-conditioned encoder–predictor Jacobian. ATR decreases when the composed map contracts along nuisance-noise directions, not necessarily when raw encoder distance alone is minimized. A useful diagnostic decomposition is

$$S_{\mathrm{comp}}(h, \mathbf{a}) = \|J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)\|_F^2 = \mathrm{tr}\left(J_E^\top J_{G_{\mathbf{a}}}^\top J_{G_{\mathbf{a}}} J_E\right),$$

with encoder-side and rollout-side reference scales

$$S_E(o) = \|J_E(o)\|_F^2, \qquad S_G(E_\theta(o), \mathbf{a}) = \|J_{G_{\mathbf{a}}}(E_\theta(o))\|_F^2.$$

Noise training can reduce the composed sensitivity through encoder-side repair, rollout-side contraction, or alignment repair: the encoder's high-sensitivity pixel directions may no longer feed the rollout map's high-gain latent directions. We use the normalized diagnostic ratio

$$\kappa(h, \mathbf{a}) = \frac{\|J_{G_{\mathbf{a}}} J_E\|_F^2}{\|J_{G_{\mathbf{a}}}\|_F^2 \|J_E\|_F^2 + \varepsilon}$$

only as an orientation for mechanism analysis, not as an angle certificate.

Combining the local radius quantile with the fixed-pool margin condition gives the diagnostic noise scale

$$\sigma^*(h, \mathbf{a}, \mathcal{A}) = \frac{\Delta_{\mathcal{A}}}{2L_J \sqrt{Q_{1-\alpha}(\sum_i \lambda_i \chi_i^2)}}.$$

This is a local diagnostic approximation, not a global robustness theorem. When L_J or the weighted-chi-square tail is unavailable, the empirical cost-drift slope $\widehat{s}_C = \widehat{Q}_q(|C_h - C_{\tilde{h}}|)/\sigma$ gives only a proxy tolerance $\widehat{\sigma}^*_{\text{cost}} = \widehat{Q}_{0.50}(\Delta_{\mathcal{A}})/(2\widehat{s}_C)$.

Thus Gaussian ACPC measures action-conditioned encoder–predictor sensitivity, not encoder invariance alone. Noise training can improve robustness by reducing same-state encoder neighborhood crossing, reducing rollout amplification along the resulting perturbation directions, or changing the alignment between encoder-sensitive directions and rollout high-gain directions. It need not reduce the rollout-side Jacobian uniformly. Small encoder shifts can still be amplified by $J_{G_{\mathbf{a}}}$, while large nuisance shifts can become harmless after rollout if the product $J_{G_{\mathbf{a}}} J_E$ is small in those directions. The selective target is therefore asymmetric: the product should be small along nuisance perturbation directions, while projected rollouts for action-relevant state differences remain separated. The task-grounded near-boundary proxy margin pass event used later is

$$M = \mathbf{1}[d_{\text{proxy diff}} > d_{\text{same state noise}} + \delta_m]\,,$$

with $\delta_m = 0$ in the reported table. It is a finite state-proxy selective-consistency criterion, not a full semantic-label coverage guarantee.

The theoretical statements in this section are intended as diagnostic orientation rather than as closed-loop control guarantees. They identify the failure modes that a paired clean/noisy diagnostic should measure: rollout-disagreement tails, clean candidate-margin caveats, and collapse of task-grounded separations. They do not assert that a low empirical ATR is a uniform bound for adaptive planning, repeated replanning, or environment-feedback stability.

Collapse caveat. Low same-state ACPC alone is insufficient. A constant encoder–predictor can make every clean and perturbed rollout identical while also mapping action-distinct states to identical projected futures. This is why SMPR is part of the reported diagnostic: ATR measures predictive stability under nuisance perturbation, and SMPR evaluates whether task-grounded separations have been preserved.

4 Study protocol

The protocol separates behavior endpoints from diagnostics. Closed-loop success under Gaussian evaluation noise establishes the behavioral phenomenon; ATR and SMPR are then computed on fixed checkpoints to localize whether recovered endpoints also satisfy the selective ACPC pattern. Training seed, checkpoint epoch, evaluation seeds, and trajectory counts are kept fixed across the main comparisons.

4.1 Models and noise intervention

LeWM [19] is an end-to-end JEPA world model trained with a latent prediction loss and SIGReg regularization for latent-space model-predictive control. PushT (2D pushing), TwoRoom (2D navigation), Reacher (2D arm), and Cube (3D manipulation) form a heterogeneous stress panel spanning different nuisance/discriminability regimes. We add per-frame Gaussian noise to the input sequence as a full-sequence augmentation: each frame is independently noised with probability $p = 1.0$, and the standard deviation is drawn from $\text{Uniform}(0, \sigma_{\max})$. This is a full-sequence input-side augmentation. We sweep $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$ and evaluate observation-only Gaussian noise with the goal image unperturbed as the primary robustness endpoint.

4.2 Evaluation and diagnostics

The canonical LeWM development grid contains 36 epoch-10 checkpoints for training seed 3072: four tasks and, for each task, a no-noise baseline plus eight noise-trained checkpoints. Each checkpoint is evaluated with three evaluation seeds (42/43/44) and 100 trajectories per seed; canonical tables report the population mean ± population std across those three evaluation seeds. A completed Gaussian training-seed replication adds independent LeWM training seeds 3073 and 3074 under the same nine-point Gaussian sweep and evaluation protocol. Together, seed 3072 plus replication seeds 3073/3074 provide three training seeds for the main Gaussian closed-loop endpoint in Figure 1.

The main results follow a two-part evidence chain. Closed-loop figures and tables report training-seed mean/std for Gaussian robustness. ATR measures whether same-state clean/noisy rollout-disagreement tails contract at recovered endpoints, and SMPR evaluates whether task-grounded near-boundary separations survive that contraction. Non-Gaussian stressor scores are auxiliary scope evaluations rather than additional core diagnostics.

5 Experiments

The experiments use additive Gaussian pixel noise as the primary controlled stressor for a representation-diagnostic question. We first show that the stressor changes closed-loop behavior. We then ask where recovery appears in the model, prioritizing cross-task ACPC/paired-rollout evidence and keeping task-specific representation measurements only as discriminability analyses. The study is a matched-stressor trace through the encoder–predictor–planner chain; checkpoint ranking, external method comparison, and broad perturbation-family transfer are outside its scope.

All results use the protocol in Section 4: LeWM-base denotes the no-noise checkpoint, and LeWM+noise denotes the eight-level $\sigma_{\max}$ sweep.

5.1 JEPA control fragility under Gaussian observation noise

Table 1 reports LeWM-base success rates under observation-only Gaussian evaluation noise. Each training-seed value is first averaged over the three evaluation seeds; the table reports mean ± population std across the three training seeds. The goal image is kept clean in every column.

Table 1: LeWM-base under observation-only Gaussian evaluation noise. Values are mean ± population std across training seeds 3072/3073/3074; each training-seed value averages evaluation seeds 42/43/44 with 100 trajectories per evaluation seed. The goal image is kept clean.

Task	eval $\sigma = 0$	eval $\sigma = 0.03$	eval $\sigma = 0.05$	eval $\sigma = 0.08$
TwoRoom	94.67 ± 0.72	92.56 ± 1.23	81.33 ± 6.77	68.78 ± 2.45
PushT	81.78 ± 4.89	53.56 ± 12.91	19.22 ± 9.12	7.22 ± 5.81
Reacher	59.22 ± 1.03	42.56 ± 2.99	33.56 ± 1.77	18.22 ± 0.42
Cube	65.22 ± 1.10	63.00 ± 1.19	46.67 ± 1.96	43.11 ± 3.27

LeWM-base is strong on unperturbed images, but observation noise alone produces large closed-loop losses at $\sigma = 0.08$: PushT loses 74.56 ± 5.55 points, Reacher 41.00 ± 1.44 points, TwoRoom 25.89 ± 2.38 points, and Cube 22.11 ± 2.78 points between the first and last columns. The progression across noise levels is smooth enough to show that the endpoint is not an isolated artifact of one severity setting. PushT degrades most sharply, consistent with contact-heavy continuous control being sensitive to visual precision.

5.2 Noise augmentation reveals matched-Gaussian recovery bands

Figure 1 shows the full LeWM Gaussian sweep aggregated across training seeds 3072/3073/3074. Each plotted point first averages evaluation seeds 42/43/44 within a training seed and then reports mean and population std across the three training seeds. The main qualitative pattern is a broad, task-dependent recovery band rather than a stable universal checkpoint ranking.

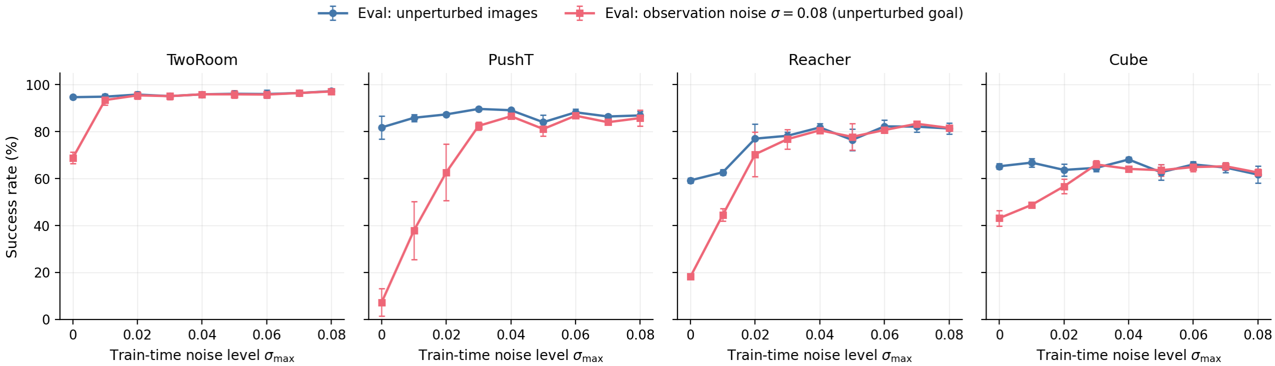

Figure 1: Three-training-seed LeWM Gaussian sweep across four tasks. Each point averages evaluation seeds 42/43/44 within each training seed and then reports mean across training seeds 3072/3073/3074; error bars show population std across training seeds. Blue circles: unperturbed evaluation; red squares: observation-only Gaussian noise $\sigma = 0.08$ with an unperturbed goal. The curves document broad task-dependent recovery bands rather than checkpoint rankings.

Because Figure 1 already aggregates the full sweep across three training seeds, we do not report a separate endpoint leaderboard in the main text. The fixed $\sigma_{\max} = 0.08$ endpoint substantially improves the observation-noise endpoint on TwoRoom, PushT, and Reacher, with a weaker positive Cube signal. Exact all-seed sweep values are reported in Appendix C for reproducibility.

A three-seed blur/resize evaluation later in this section serves as an auxiliary severe-stressor evaluation outside the matched-Gaussian endpoint. The next subsection asks whether the recovered Gaussian endpoints also move in the ATR/SMPR diagnostics.

5.3 ATR/SMPR selective-ACPC diagnostics

For the main diagnostic, ATR is the 90th percentile of same-state clean/noisy ACPC-H rollout distance normalized by the clean transition scale. SMPR is the task-grounded near-boundary selective-margin pass rate at margin zero. Both are computed without retraining on the fixed no-noise and $\sigma_{\max} = 0.08$ checkpoints for training seeds 3072/3073/3074. Table 2 reports population mean/std across training seeds.

Table 2: ATR/SMPR selective-ACPC diagnostics over training seeds 3072/3073/3074. ATR is the q90 normalized same-state clean/noisy action-conditioned rollout disagreement; lower is better. SMPR is the task-grounded near-boundary selective-margin pass rate; higher is better.

Task	ATR base	ATR std0.08	SMPR base	SMPR std0.08
TwoRoom	1.509 ± 0.010	0.111 ± 0.016	0.34 ± 0.05	0.99 ± 0.01
PushT	3.580 ± 0.260	0.247 ± 0.017	0.44 ± 0.17	1.00 ± 0.00
Reacher	2.628 ± 0.048	0.082 ± 0.005	0.73 ± 0.03	1.00 ± 0.00
Cube	2.320 ± 0.045	0.100 ± 0.007	0.45 ± 0.03	1.00 ± 0.00

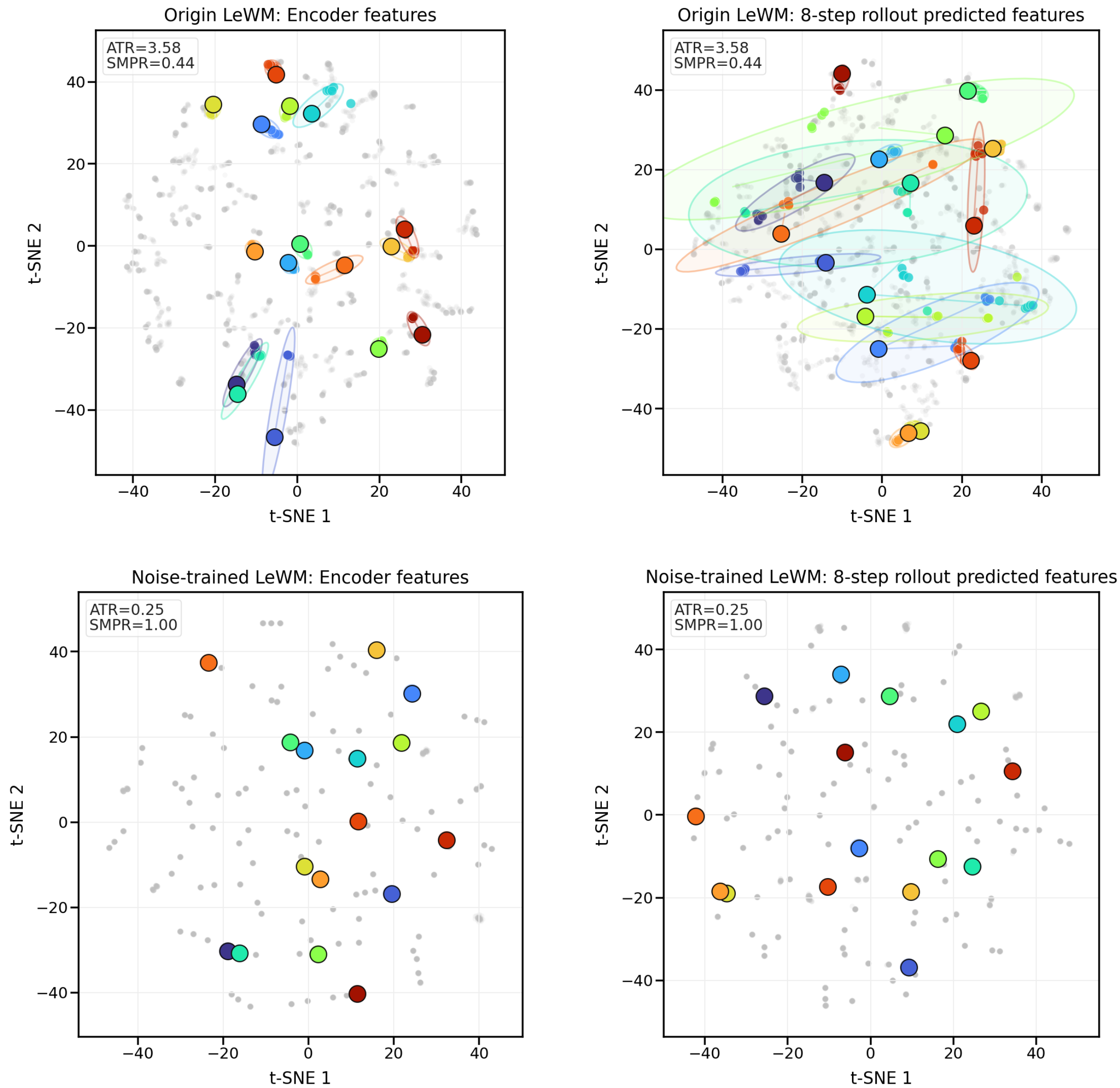

Figure 2: Qualitative PushT ACPC neighborhood t-SNE visualization for the Origin LeWM versus $\sigma_{\max} = 0.08$ noise-trained contrast. The four panels show encoder features and 8-step rollout predicted features for repeated same-state perturbation views; colored clusters mark fixed random anchors. The t-SNE projection is qualitative. ATR and SMPR are the quantitative diagnostics and are computed in the original projected-rollout diagnostic space.

The two diagnostics move in the intended selective direction on all four tasks: recovered endpoints have much smaller same-state predictive tails and much higher task-grounded margin pass rates. Figure 2 provides qualitative visual intuition on PushT: repeated noisy views of the same state become tighter after noise training in both encoder features and 8-step rollout predicted-feature projections. We interpret the post-rollout tightening as the planner-facing consequence of a lower composed encoder–rollout response to actual noise-induced perturbations, not as evidence that the rollout-side Jacobian uniformly contracts. The quantitative claim remains on ATR, SMPR, and closed-loop behavior rather than distances in the t-SNE plane. Cube remains the useful boundary case: ATR and SMPR improve strongly, but the closed-loop Gaussian recovery in Figure 1 is weaker than PushT or Reacher, so closed-loop evaluation remains the behavioral authority.

SMPR uses programmatic task-state proxy labels, not oracle semantic annotations: TwoRoom uses doorway/room-side labels, PushT uses T-block pose cells, Reacher uses target/end-effector relation bins, and

Cube uses cube-pose/goal-relation cells. Each anchor selects the closest label-crossing neighbor inside the closest 35% state-distance neighborhood and passes if that clean different-state rollout distance exceeds the same-state noisy radius. The no-noise baselines are weak under this stricter criterion, while the high-noise endpoint reaches near-one pass rate. The joint fixed-pool top-1 flip guard mitigates this proxy-label limitation by evaluating whether candidate rankings stabilize, but it does not replace stronger oracle-level contact, topology, action-value, or cost-to-go semantics, especially for PushT and Cube.

5.4 Full-sweep diagnostic separation and held-out validation

Endpoint diagnostics are supplemented by a training-free full-sweep analysis over all $4 \times 3 \times 9 = 108$ task–training-seed–checkpoint rows. Figure 3 overlays the closed-loop observation-noise score with normalized ATR tail, SMPR failure $(1 - \text{SMPR})$, and fixed-pool top-1 disagreement. The recovered portions of the Gaussian sweep consistently occupy lower normalized ATR and higher SMPR than fragile rows. We also recompute the fixed 65-candidate pool from checkpoints for all 108 rows, keeping sample-level maximum cost drift and observed top-1 flips. In Table 9, recovered rows have higher sufficient-event pass rates and much lower fixed-pool top-1 flip rates, while strict q10/q95 gaps remain negative. Table 10 turns the sample-level recomputation into empirical fixed-pool risk calibration by reporting Wilson intervals for cert-pass, top-1 flip, and top-1 flip conditional on cert-pass. This treats q10/q95 as a conservative negative calibration result and uses paired event rates, rather than a tuned quantile gap, as the fixed-pool evidence. The evidence remains a fixed-pool empirical analysis, not a calibrated theorem probability bound.

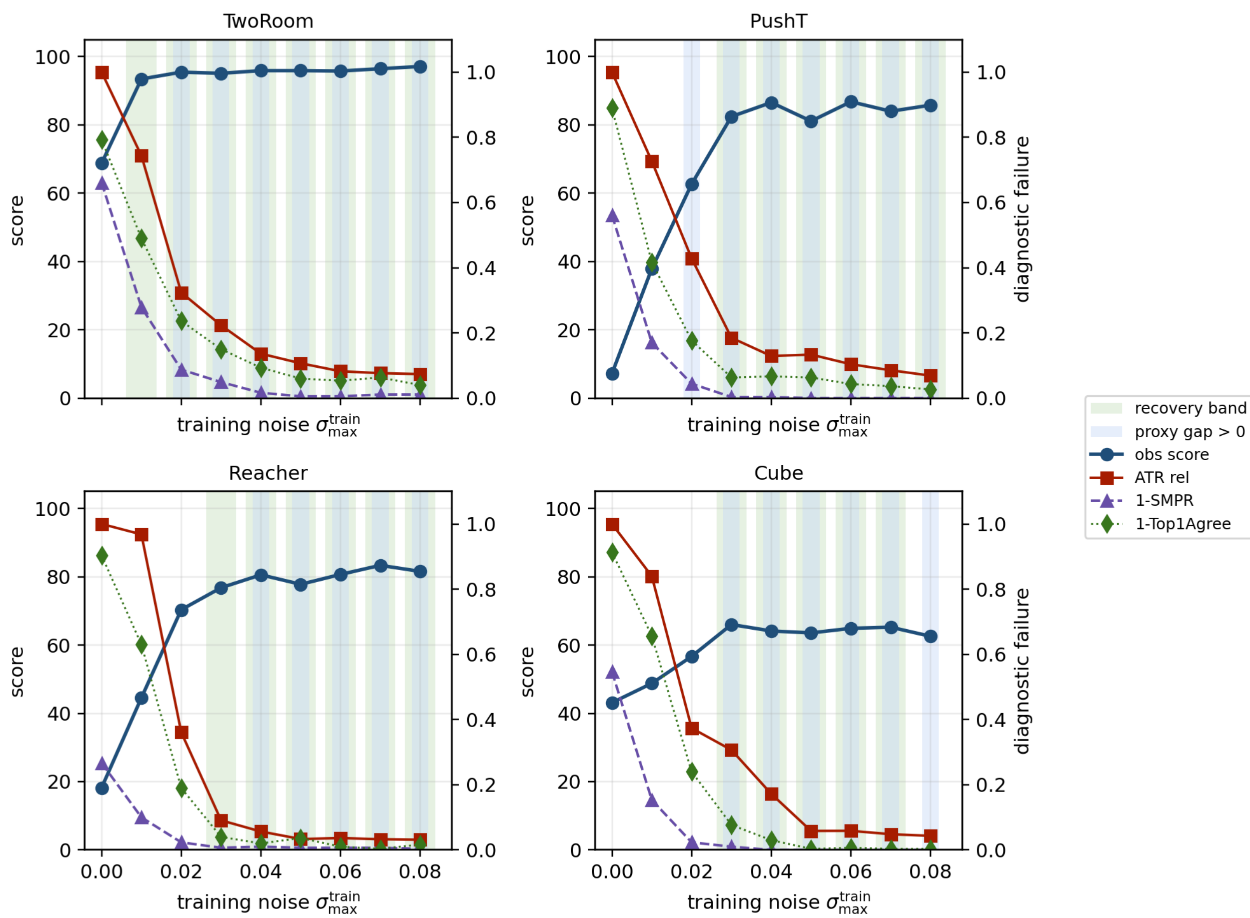

Figure 3: Full-sweep diagnostic separation across the Gaussian training grid. Each task panel averages training seeds 3072/3073/3074 at the same $\sigma_{\max}$ value. Green bands mark behavioral recovery-band points; blue bands mark positive summary-level fixed-pool proxy gaps. The right-axis curves show normalized ATR tail and SMPR failure $(1 - \text{SMPR})$, plus fixed-pool top-1 disagreement; lower is better for all three diagnostic-failure curves.

We also freeze simple diagnostic gates on calibration units before evaluating held-out units. In leave-one-training-seed-out validation, gates calibrated on two seeds and evaluated on the held-out seed have mean absolute recovery-onset error 0.007 in $\sigma_{\max}$ units, maximum error 0.020, precision 0.90, and recall 0.91. In leave-one-task-out validation, the corresponding values are 0.008, 0.020, 0.90, and 0.88. These held-out validations support ATR/SMPR as matched-Gaussian recovered-vs-fragile diagnostics, not as universal checkpoint rankers; task-dependent boundary errors remain visible.

Table 3: Held-out diagnostic-region validation. Gates are selected on calibration seeds or tasks and evaluated on held-out sweeps. Start error is in $\sigma_{\max}^{\text{train}}$ units; positive means the diagnostic starts later than the behavioral recovery band.

Split	blocks	mean |start err|	max |start err|	precision	recall
seed held-out	12	0.007	0.020	0.90	0.91
task held-out	12	0.008	0.020	0.90	0.88

Threshold sensitivity is reported in Appendix B.3. Across recovery fractions 0.70/0.80/0.90 and clean-score tolerances of 3/5/10 points, all four tasks keep the expected recovered-vs-fragile direction: recovered rows have lower median normalized ATR and higher median SMPR than fragile rows. Table 12 keeps the interpretation joint: ATR is the radius term, while SMPR and fixed-pool top-1 flip are guard-side criteria and are not standalone robustness metrics. ATR q80/q95 and positive-margin SMPR variants remain unavailable in the current full-sweep ATR/SMPR summaries rather than inferred.

5.5 Fixed-pool radius–margin calibration

The radius–margin theory predicts that a checkpoint should remain diagnostically stable under the matched stressor when the high-tail same-state predictive radius is contained within the relevant planner or task-discriminability margin. We therefore add a training-free full-sweep validation using existing three-training-seed evaluation and diagnostic summaries. The behavioral recovery band is defined only from closed-loop observation-noise scores: a checkpoint is marked as inside the recovery band if its observation-only $\sigma = 0.08$ success is at least 80% of the way from the no-noise baseline to the maximum grid value for that task, while clean success stays within 5 points of the no-noise clean score. The diagnostic proxy is computed independently from fixed-pool diagnostic summaries as

$$\widehat{\Gamma}_{\rho,0.08} = \widehat{Q}_{0.50}(\Delta_{\mathcal{A}}) - 2\widehat{Q}_{0.90}(|C_h - C_{\tilde{h}}|).$$

A positive proxy gap is the available summary-level counterpart of the fixed-pool margin condition. The summary-level overlay is kept as a mechanism proxy, not a planner-margin certificate: it uses median clean candidate margin and q90 paired cost drift. The separate sample-level recomputation evaluates the sufficient event directly over the fixed 65-candidate pool for all 108 rows and also reports strict q10/q95 gaps. Those q10/q95 gaps remain negative even on recovered rows, indicating that q10/q95 is too conservative for the current fixed-pool cost scale and should not be used as the paper's claimed threshold. The reported fixed-pool evidence is event-rate based: across all tasks, median cert-pass rises from 0.06 on fragile rows to 0.61 on recovered rows, median top-1 flip falls from 0.54 to 0.04, and Table 10 reports Wilson intervals for $\hat{p}_{\text{cert}}$, $\hat{p}_{\text{flip}}$, and $\hat{p}_{\text{flip}|\text{cert}}$. In the sampled anchors, observed top-1 flips conditioned on cert-pass are zero in both fragile and recovered splits, so the sufficient event is empirically conservative rather than overfit. Thus this analysis instantiates the radius–margin mechanism as empirical fixed-pool risk calibration, not as a calibrated probability certificate or checkpoint-ranking rule. Because the Gaussian sweep uses equal $\sigma_{\max}$ increments while the tasks have unequal visual and dynamical sensitivity, aggregate grid-point F1 or interval IoU is not used as the primary validation criterion. Appendix B.1 reports boundary-aware recovery/proxy alignment, and Appendix B.2 reports the empirical risk-calibration details.

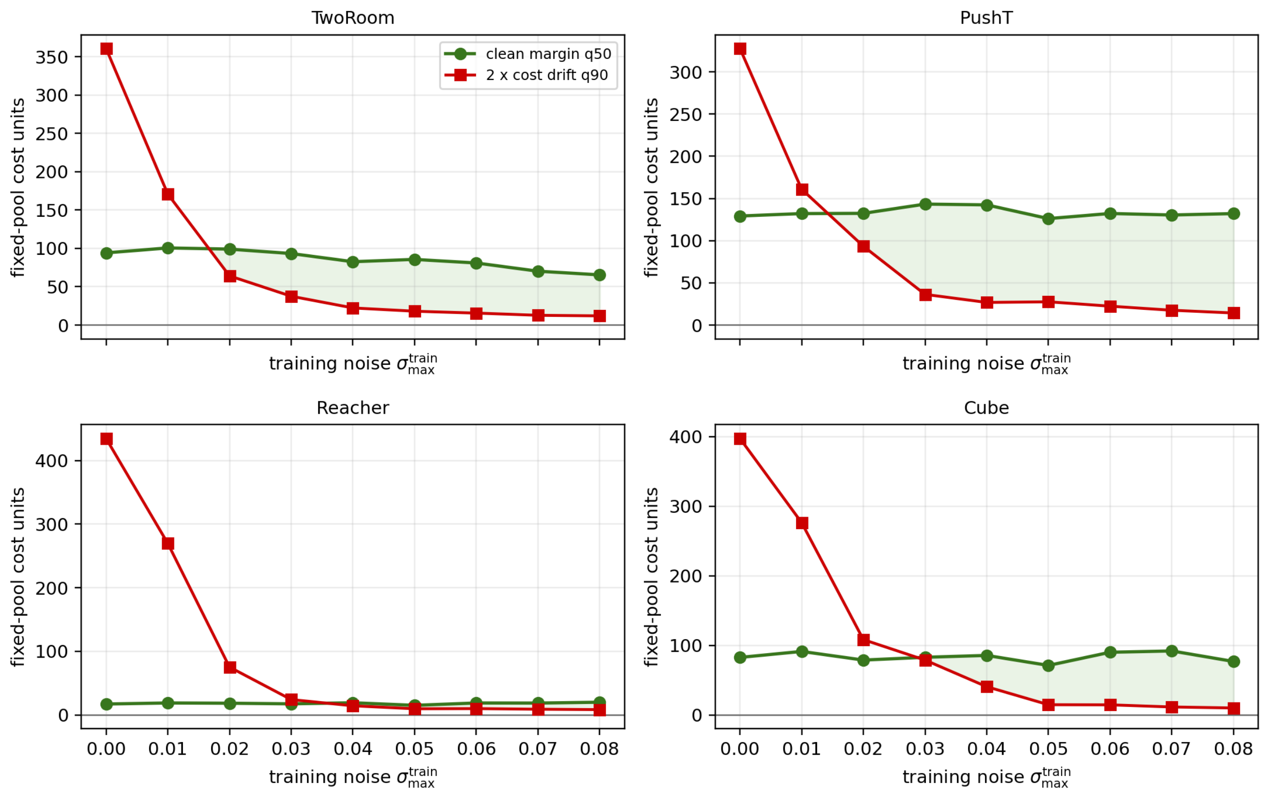

Figure 4: Fixed-pool radius–margin overlap proxy across the full Gaussian training sweep. Curves use the available summary-level fixed-pool summaries: q50 clean candidate margin and twice the q90 paired cost drift. The separate sample-level recomputation in Table 9 provides q10/q95 gaps and event rates; q10/q95 gaps remain negative, so the analysis supports the mechanism rather than a calibrated planner-margin certificate. The sampled pool is fixed rather than adaptive CEM.

The summary-level fixed-pool overlap proxy aligns with the broad recovery band but does not exactly reproduce it. TwoRoom and PushT cross the proxy gap within one grid step of the recovery-band onset, Cube aligns on the discrete grid, and Reacher is conservative by two grid steps. This is the expected level of evidence for a fixed-pool diagnostic analysis: it explains the radius–margin mechanism behind the observed recovery band without replacing closed-loop evaluation. The Reacher mismatch is informative rather than hidden: fixed-pool top-1 agreement already rises at the behavioral recovery onset, while the summary-level cost-margin proxy becomes positive later. To keep the fixed-pool cost-margin proxy separate from a checkpoint-selection claim, we do not turn this proxy into a radius–margin ATR/SMPR gate table. The full-sweep ATR/SMPR held-out validation in Section 5.4 handles recovered-vs-fragile diagnostic separation; this subsection isolates the summary-level fixed-pool cost-margin mechanism and its limits.

5.6 Local Gaussian sensitivity analysis

To evaluate the local Gaussian sensitivity interpretation without claiming a global guarantee, we run two checkpoint-local analyses on the base, recovery-onset, and endpoint checkpoints. First, a finite-difference analysis perturbs the same observation histories with small Gaussian noise levels and estimates $\mathbb{E}[R_\sigma^2]/\sigma^2$ from the action-conditioned rollout radius. Table 13 reports a consistent endpoint/base reduction in this local slope, from 0.002 to 0.008 of the no-noise-checkpoint value across tasks, alongside the ATR contraction. Second, Table 14 estimates exact-autograd JVP/Hutchinson Frobenius traces for the encoder map, the rollout map, and their composition. The composed trace falls sharply at the endpoint on all tasks (endpoint/base ratios 0.003, 0.010, 0.006, and 0.028 for TwoRoom, PushT, Reacher, and Cube). In this sweep, the decomposition attributes this mainly to encoder-side sensitivity reduction; the rollout-side trace is task-dependent rather than uniformly smaller, so we do not claim a standalone predictor-Jacobian repair. This distinction also clarifies the qualitative ACPC-basin plot: tighter post-rollout feature clouds measure the composed response to the actual perturbation directions, not J_G alone. These analyses support the composed encoder–predictor sensitivity explanation while remaining local mechanism evidence rather than closed-loop robustness proofs.

5.7 Evaluation under bounded non-Gaussian stressors

The main evidence remains the matched Gaussian-noise endpoint. After freezing the Gaussian grid, we ran strongest-severity non-Gaussian stressor evaluations to test how the fixed $\sigma_{\max} = 0.08$ endpoint transfers outside the training perturbation family. Table 4 reports the non-Gaussian behavior together with stressor-specific ATR/SMPR diagnostics recomputed under the same blur or resize perturbation. TwoRoom and Reacher improve under strongest blur, whereas PushT and Cube do not show a clear resize gain at the scale of training-seed and evaluation variance. We treat this as bounded behavior outside the matched Gaussian setting.

Table 4: Evaluation under bounded non-Gaussian stressors with stressor-specific ATR/SMPR diagnostics. Score values are mean ± population std across LeWM training seeds 3072/3073/3074; each seed point averages evaluation seeds 42/43/44 from strongest-severity non-Gaussian runs. The ATR and SMPR columns report raw no-noise → noise-trained diagnostic values under the row stressor; lower ATR and higher SMPR are better.

Task	evaluation stressor	no-noise score	noise-trained score	ATR	SMPR
TwoRoom	blur $k = 15$	47.67 ± 5.44	90.78 ± 5.38	$1.61 \to 1.24$	$0.16 \to 0.77$
Reacher	blur $k = 15$	22.00 ± 3.78	71.22 ± 1.10	$2.81 \to 0.54$	$0.60 \to 0.98$
PushT	resize 0.25	63.44 ± 14.05	66.33 ± 8.38	$1.77 \to 1.53$	$0.93 \to 0.96$
Cube	resize 0.25	57.00 ± 1.96	56.11 ± 0.57	$1.35 \to 1.59$	$0.98 \to 0.95$

The raw unseen diagnostics co-move with behavior at this severe-stressor scale. The blur rows combine large behavior gains with lower ATR and higher SMPR after noise training; PushT resize has only small score and diagnostic movement; Cube resize is neutral to slightly negative on behavior and also moves in the wrong diagnostic direction. These rows therefore support only a bounded severe-stressor association rather than a general perturbation-transfer claim.

6 Discussion and limitations

What the evidence supports. This is a controlled diagnostic study, not a method paper. Closed-loop evaluation establishes the behavior; ATR measures whether same-state noisy views have small action-conditioned predictive tails; SMPR evaluates whether task-grounded near-boundary distinctions survive that contraction. Full-sweep and held-out validations support recovered-vs-fragile diagnostic separation under the matched Gaussian stressor. The radius–margin analysis adds fixed-pool cost-margin and sample-level event-rate evidence over the Gaussian grid, and the finite-difference and exact-JVP/Hutchinson analyses evaluate the local Gaussian sensitivity mechanism; mismatches are reported as diagnostic limits rather than hidden by the endpoint result. The primary statistic is the full three-training-seed Gaussian replication, with evaluation-seed variance kept as the within-seed measurement scale. These standard deviations are descriptive over three training seeds, not asymptotic significance claims.

Scope. Gaussian pixel noise remains the training/evaluation axis. The blur/resize rows are bounded behavior outside the matched Gaussian setting, not a transfer theorem. The radius–margin certificate is fixed-pool and matched-perturbation only: it does not cover adaptive CEM resampling, repeated replanning, or environment-feedback trajectory guarantees. The Gaussian sensitivity analyses are local: finite-difference slopes and exact-JVP/Hutchinson trace estimates do not provide a global robustness or closed-loop guarantee. The empirical fixed-stressor diagnostic region is an analysis view over matched checkpoints rather than a closed-loop guarantee. ATR/SMPR are not a checkpoint-ranker for $\sigma_{\max}$ and do not replace closed-loop evaluation inside a broad recovery band; SMPR is only a guard-side component of the joint diagnostic and uses programmatic proxy labels rather than oracle action-value semantics. The fixed-pool top-1 flip guard evaluates planning consistency but does not replace stronger semantic labels for contact- or pose-sensitive tasks. We do not claim superiority over DrQ-style augmentation, TD-MPC2, DreamerV3, or robust MPC methods; the contribution is a no-retraining diagnostic for JEPA latent world-model checkpoints under a controlled Gaussian stressor.

What remains. The results argue against a too-strong reading of latent prediction: future-latent prediction alone does not guarantee closed-loop visual robustness. The joint guard-side validation shows that the low-ATR region also preserves SMPR and fixed-pool candidate stability, but stronger semantic discriminability evidence would still require additional labels. Hand-labeled or simulator-derived contact, topology, action-value, or cost-to-go labels remain future validation, especially for PushT and Cube. Future methods can turn ATR/SMPR into objectives, but this paper keeps that as a separate method-paper path after diagnostic validity is established.

7 Conclusion

This paper frames Gaussian-noise robustness in JEPA latent world-model control as a selective diagnostic problem. Closed-loop behavior establishes the phenomenon; ATR measures same-state predictive-stability tails; SMPR evaluates task-grounded anti-collapse margins. On fixed LeWM checkpoints, input-side noise training recovers broad matched-Gaussian recovery bands across three training seeds. Across the full Gaussian training sweep, recovered rows occupy low-ATR/high-SMPR regions, held-out seed/task validations support recovered-vs-fragile diagnostic separation, and full-sweep fixed-pool event-rate recomputation links the radius–margin mechanism to candidate stability. Finite-difference and exact-JVP/Hutchinson analyses further connect ATR contraction to lower local composed encoder–rollout sensitivity, while keeping closed-loop behavior as the authority.

The theory is deliberately diagnostic: sampled-pool ACPC exposes why tail risk matters and why clean margins prevent closed-loop guarantees, while local Gaussian linearization connects the measured tail to action-conditioned encoder–predictor sensitivity. The resulting view is a radius–margin diagnostic theory for fixed-checkpoint Gaussian robustness: ATR estimates the perturbation-tube radius tail, SMPR estimates margin preservation, and their conjunction defines the empirical fixed-stressor diagnostic region studied here. The broader implication is not solved generalization; it is a concrete evaluation target for future methods: stable action-conditioned predictive dynamics under nuisance variation, while preserving distinctions needed for planning.

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A Proofs for ACPC Diagnostics

These proofs support the diagnostic use of ATR and SMPR; they do not provide closed-loop control guarantees.

Proposition 2 (ACPC controls candidate-cost drift)**.** *Assume that J is locally L_J-Lipschitz on the clean/perturbed projected-rollout neighborhood evaluated by the fixed candidate set under metric d_H. If*

$$d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))\big) \leq \epsilon,$$

then

$$|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \leq L_J\epsilon.$$

Proof. The planner cost J is L_J-Lipschitz on the projected-rollout sequence, so the cost difference is bounded by L_J times the clean/perturbed projected-rollout distance. □

Proposition 3 (Candidate top-1 stability)**.** *Let $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\}$ be the shared candidate set. Assume lower cost is preferred, and let $\mathbf{a}^{(1)}$ and $\mathbf{a}^{(2)}$ denote the lowest- and second-lowest-cost candidates on the clean branch. Suppose that for every candidate j,*

$$|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq \eta.$$

If the clean branch has a strict top-1/top-2 margin

$$\Delta = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g) > 2\eta,$$

then the perturbed branch selects the same top-1 candidate.

Proof. For any candidate $\mathbf{a}^j \neq \mathbf{a}^{(1)}$,

$$C_{\tilde{h}}(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^{(1)}, g) \geq C_h(\mathbf{a}^j, g) - C_h(\mathbf{a}^{(1)}, g) - 2\eta \geq \Delta - 2\eta > 0.$$

Thus every non-top candidate remains more costly than the clean top-1 candidate on the perturbed branch. □

Proof of Theorem 1.

Proof. Let $\mathcal{B} = \{\max_{1\leq j\leq K} D_j \leq \epsilon\}$. Since each candidate is drawn from q and a single draw has tail probability at most α, the union bound gives

$$\Pr_{\mathcal{A}}(\mathcal{B}^c) \leq \sum_{j=1}^{K} \Pr[D_j > \epsilon] \leq K\alpha.$$

On $\mathcal{B}$, local Lipschitzness of the planner cost gives $|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq L_J\epsilon$ for every candidate. By the top-1 stability proposition, the perturbed branch can choose a different top candidate on $\mathcal{B}$ only when $\Delta_{\mathcal{A}} \leq 2L_J\epsilon$. Therefore the top-1 flip event is contained in

$$\mathcal{B}^c \cup \{\Delta_{\mathcal{A}} \leq 2L_J\epsilon\},$$

which yields the fixed-pool instability bound without any independence assumption beyond the single-candidate tail statement. For the selective diagnostic statement, let $\mathcal{D} = \{M_{\text{diff}} \leq R_\sigma + \delta_m\}$. Selective failure is contained in the union of the fixed-pool planning-instability event and $\mathcal{D}$. Applying the union bound again and using $\Pr(\mathcal{D}) \leq \gamma$ gives the final bound. □

Local Gaussian sensitivity. For small Gaussian pixel noise ξ, a first-order expansion gives $E_\theta(o + \xi) = E_\theta(o) + J_E(o)\xi + \mathcal{R}(\xi)$ with $\|\mathcal{R}(\xi)\| = O(\|\xi\|^2)$. Composing the encoder with the action-conditioned rollout map shows that same-state rollout disagreement measures local sensitivity of the encoder–predictor composition, not encoder invariance alone. ATR reports a high-quantile version of that paired rollout disagreement.

B Experimental Details

These details define the fixed evaluation and diagnostic protocol used by the main tables.

LeWM is evaluated on PushT, TwoRoom, Reacher, and Cube using the canonical three evaluation seeds 42/43/44 and 100 trajectories per seed. The main Gaussian robustness endpoint perturbs observation pixels with Gaussian noise at standard deviation 0.08 while keeping the goal image clean. Noise-trained checkpoints use full-sequence input-side Gaussian augmentation with $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$.

ATR is computed at each task–training-seed–checkpoint row as the 90th percentile same-state ACPC-H distance normalized by the clean transition scale. SMPR uses task-grounded near-boundary proxy labels at margin zero. The paper reports population mean/std across training seeds 3072/3073/3074.

B.1 Radius–margin parameter interpretation

Theorem 1 separates the quantities needed for a numerical fixed-pool certificate from the lower-sample diagnostic summaries reported in the experiments. In the fixed-pool diagnostic summaries used for Figures 3 and 4, the candidate count is $K = 65$ for every task–checkpoint row. The theorem's single-candidate tail probability α is not fixed by the q90 reporting level used by ATR. A q90 summary descriptively leaves 10% of measured same-state radii above the reported quantile, but that descriptive exceedance rate is not the α used in the theorem. If it were naively inserted into the union-bound term, the result would be $65 \times 0.1 = 6.5$, which is vacuous. A non-vacuous numerical candidate-flip bound would require a much smaller single-candidate tail probability or a different concentration argument. We therefore use q90 ATR and the available q50/q90 cost-margin overlap as diagnostic evidence for the radius–margin mechanism, not as a calibrated probability guarantee.

Table 5: Fixed-pool radius–margin interpretation used in the diagnostic analysis. The behavioral recovery band is defined from closed-loop observation-only $\sigma = 0.08$ scores; the proxy-positive range uses $\widehat{Q}_{0.50}(\Delta_{\mathcal{A}}) - 2\widehat{Q}_{0.90}(|C_h - C_{\tilde{h}}|) > 0$ on the available summary-level fixed-pool summaries. These ranges explain the recovery/proxy bands in Figure 3 and the overlap proxy in Figure 4; they are not used to claim a calibrated flip-probability certificate.

Task	K	recovery band	proxy-positive range	interpretation
TwoRoom	65	0.01–0.08	0.02–0.08	proxy misses the first recovery-band checkpoint
PushT	65	0.03–0.08	0.02–0.08	proxy includes one earlier checkpoint
Reacher	65	0.02–0.08	0.04–0.08	proxy is conservative at low noise-training levels
Cube	65	0.03–0.08	0.03–0.08	proxy and recovery-band ranges align

The one-grid tolerance is $\tau_{\text{grid}} = 0.01$ in $\sigma_{\max}$ units. Start-boundary error is defined as $\rho^{\text{diag}}_{\text{start}} - \rho^{\text{beh}}_{\text{start}}$, so positive values mean the diagnostic proxy is later and negative values mean it fires earlier.

Table 6: Boundary-aware interpretation of the fixed-pool radius–margin proxy. Equal spacing in $\sigma_{\max}$ is not equal spacing in task sensitivity, so these rows are interpreted as task-conditioned boundary comparisons rather than grid-point classifier scores.

Task	recovery band	proxy range	start error	within τ_{grid}?	interpretation
TwoRoom	0.01–0.08	0.02–0.08	+0.01	yes	proxy misses the first recovery-band checkpoint
PushT	0.03–0.08	0.02–0.08	−0.01	yes	proxy fires one grid step early
Reacher	0.02–0.08	0.04–0.08	+0.02	partial	fixed-pool cost-margin proxy is conservative
Cube	0.03–0.08	0.03–0.08	0.00	yes	proxy and recovery band align

The measured fixed-pool flip rate is the clean/noisy top-1 disagreement rate under the same candidate set, so one minus this flip rate gives an empirical top-1 agreement analysis that is closer to the theorem's intermediate statement than closed-loop score. It still does not cover adaptive CEM resampling or repeated replanning.

Table 7: Fixed-pool top-1 agreement analysis derived from the measured fixed-pool flip rate. Rows show the no-noise baseline, the first closed-loop recovery-band checkpoint on the discrete sweep, and the matched $\sigma_{\max} = 0.08$ endpoint.

Task	checkpoint	cost drift q90	clean margin q50	proxy gap	Top1Agree
TwoRoom	base 0.00	180.33	94.14	−266.52	0.207
TwoRoom	onset 0.01	85.42	100.56	−70.28	0.510
TwoRoom	endpoint 0.08	6.00	65.46	53.46	0.960
PushT	base 0.00	163.82	129.21	−198.44	0.110
PushT	onset 0.03	18.16	143.29	106.96	0.937
PushT	endpoint 0.08	7.15	132.05	117.74	0.973
Reacher	base 0.00	217.30	17.29	−417.32	0.097
Reacher	onset 0.02	37.67	18.54	−56.80	0.810
Reacher	endpoint 0.08	4.24	20.08	11.59	0.983
Cube	base 0.00	198.70	82.76	−314.65	0.087
Cube	onset 0.03	39.48	82.95	3.99	0.923
Cube	endpoint 0.08	5.05	76.94	66.85	0.997

B.2 Empirical fixed-pool risk calibration

For each sampled state h_i, let $E_{\text{pool}}(h_i)$ be the fixed-pool sufficient event that the maximum paired candidate-cost drift over the $K = 65$ fixed candidates is below one half of the clean top-1/top-2 margin. Let $F_{\text{pool}}(h_i)$ be the observed clean/noisy fixed-pool top-1 disagreement. We report three empirical rates over the sampled anchors in each split:

$$\begin{aligned}
\hat{p}_{\text{cert}} &= \frac{1}{n}\sum_i \mathbf{1}[E_{\text{pool}}(h_i)],\\
\hat{p}_{\text{flip}} &= \frac{1}{n}\sum_i \mathbf{1}[F_{\text{pool}}(h_i)],\\
\hat{p}_{\text{flip}|\text{cert}} &= \frac{\sum_i \mathbf{1}[F_{\text{pool}}(h_i) \wedge E_{\text{pool}}(h_i)]}{\sum_i \mathbf{1}[E_{\text{pool}}(h_i)]}.
\end{aligned} \tag{11}$$

The rates are computed over the full 108-row Gaussian sweep after splitting checkpoints by the closed-loop recovery-band label, and Table 10 reports Wilson 95% intervals. The strict q10/q95 gap remains negative even at recovered checkpoints; this is a useful negative calibration result showing that the q10/q95 rule is too conservative for the current fixed-pool cost scale. We therefore use the paired event-rate calibration, especially $\hat{p}_{\text{flip}|\text{cert}}$, as the reported fixed-pool evidence instead of tuning a quantile threshold.

In this interpretation, β_{plan} and β_{disc} name empirical failure components rather than calibrated probabilities. The planning component is represented by the fixed-pool cost-margin proxy, top-1 agreement analysis, and the sample-level event-rate recomputation; the discriminability component is represented by SMPR, equivalently its failure rate $1 - \text{SMPR}$. The closed-loop recovery band explains the fixed-pool radius–margin validation rather than converting the separate ATR/SMPR diagnostic-region validation into a threshold-classifier claim.

B.3 Supplementary diagnostic analyses

The held-out validation in Table 3 is supported by supplementary fixed-pool, sample-level, sensitivity, and threshold-sensitivity analyses. Table 8 reports full-sweep top-1 agreement and proxy-gap rows selected from the available summary-level fixed-pool summaries. Table 9 adds the full-sweep sample-level event-rate recomputation, and Table 10 reports the empirical fixed-pool risk calibration: cert-pass, top-1 flip, and top-1 flip conditional on cert-pass with Wilson intervals. These intervals quantify estimation uncertainty, not calibrated theorem probabilities. Table 11 preserves the base/endpoint view. Table 12 reports ATR together with SMPR and fixed-pool top-1 flip to keep the guard-side interpretation joint rather than standalone. Table 13 reports the finite-difference local Gaussian sensitivity analysis, and Table 14 reports the exact-JVP/Hutchinson local trace decomposition. Table 15 evaluates whether recovered-vs-fragile ATR/SMPR direction is stable under reasonable behavioral-label thresholds; unavailable ATR/SMPR tail variants are reported as unavailable rather than inferred.

Table 8: Summary-level fixed-pool top-1 agreement analysis. Top1Agree is derived from the measured fixed-pool flip rate; cert-pass rates require sample-level maximum cost-drift traces.

Task	row	$\sigma_{\max}^{\text{train}}$	Top1Agree	proxy gap	proxy pass rate
TwoRoom	base	0.00	0.207	-266.52	0.00
TwoRoom	proxy onset	0.02	0.763	34.93	1.00
TwoRoom	endpoint	0.08	0.960	53.46	1.00
PushT	base	0.00	0.110	-198.44	0.00
PushT	proxy onset	0.02	0.823	38.95	0.67
PushT	endpoint	0.08	0.973	117.74	1.00
Reacher	base	0.00	0.097	-417.32	0.00
Reacher	proxy onset	0.04	0.980	4.79	1.00
Reacher	endpoint	0.08	0.983	11.59	1.00
Cube	base	0.00	0.087	-314.65	0.00
Cube	proxy onset	0.03	0.923	3.99	0.67
Cube	endpoint	0.08	0.997	66.85	1.00

Table 9: Full-sweep sample-level fixed-pool event-rate calibration. Rows split all task–seed–$\sigma_{\max}$ checkpoints by the closed-loop recovery-band label. Cert-pass is the fraction of sampled states satisfying the fixed-pool sufficient event; top-1 flip is the observed clean/noisy fixed-pool best-candidate disagreement. These event rates strengthen the mechanism analysis but are not calibrated probability bounds.

Task	cert-pass fragile → recovered	top-1 flip fragile → recovered	q10/q95 gap fragile → recovered
TwoRoom	0.05 → 0.67	0.79 → 0.07	-1053.7 → -269.0
PushT	0.16 → 0.67	0.38 → 0.06	-797.6 → -193.5
Reacher	0.06 → 0.54	0.54 → 0.02	-873.0 → -29.3
Cube	0.08 → 0.78	0.53 → 0.01	-713.4 → -55.3
ALL	0.06 → 0.61	0.54 → 0.04	-847.7 → -136.6

Table 10: Empirical fixed-pool risk calibration. Each cell reports fragile → recovered rate [95% Wilson interval] over sampled fixed-pool anchors. The conditional column restricts the denominator to cert-pass anchors, directly evaluating whether the sufficient event is associated with low fixed-pool candidate-flip risk. These intervals quantify event-rate estimation uncertainty and are not calibrated theorem probability bounds.

Task	cert-pass	top-1 flip	flip | cert-pass
TwoRoom	0.09 [0.06,0.12] → 0.61 [0.59,0.63]	0.72 [0.68,0.77] → 0.10 [0.08,0.11]	0.00 [0.00,0.10] → 0.00 [0.00,0.00]
PushT	0.18 [0.16,0.21] → 0.65 [0.63,0.67]	0.52 [0.49,0.55] → 0.06 [0.05,0.07]	0.00 [0.00,0.02] → 0.00 [0.00,0.00]
Reacher	0.11 [0.09,0.13] → 0.52 [0.50,0.54]	0.55 [0.51,0.58] → 0.03 [0.02,0.04]	0.00 [0.00,0.04] → 0.00 [0.00,0.00]
Cube	0.32 [0.29,0.35] → 0.64 [0.62,0.67]	0.47 [0.44,0.50] → 0.04 [0.03,0.05]	0.00 [0.00,0.01] → 0.00 [0.00,0.00]
ALL	0.20 [0.19,0.21] → 0.61 [0.59,0.62]	0.53 [0.52,0.55] → 0.06 [0.05,0.06]	0.00 [0.00,0.01] → 0.00 [0.00,0.00]

Table 11: Sample-level fixed-pool endpoint calibration. Cert-pass is the fraction of sampled states where the maximum paired candidate-cost drift over the fixed 65-candidate pool is below half the clean top-1/top-2 margin. Values are mean $\pm$ population std across training seeds 3072/3073/3074 with 100 sampled states per seed. Strict q10/q95 gaps remain negative, so these rows support a fixed-pool mechanism analysis rather than a calibrated probability certificate.

Task	cert-pass base → std0.08	top-1 flip base → std0.08	q10/q95 gap base → std0.08
TwoRoom	$0.05 \pm 0.00 \to 0.73 \pm 0.03$	$0.80 \pm 0.02 \to 0.03 \pm 0.00$	$-1084.6 \pm 44.0 \to -179.0 \pm 44.2$
PushT	$0.01 \pm 0.01 \to 0.74 \pm 0.04$	$0.92 \pm 0.03 \to 0.03 \pm 0.02$	$-917.4 \pm 44.3 \to -126.2 \pm 13.5$
Reacher	$0.00 \pm 0.00 \to 0.61 \pm 0.01$	$0.88 \pm 0.03 \to 0.01 \pm 0.00$	$-935.2 \pm 32.1 \to -24.7 \pm 2.3$
Cube	$0.00 \pm 0.00 \to 0.83 \pm 0.01$	$0.92 \pm 0.03 \to 0.00 \pm 0.00$	$-1005.3 \pm 13.9 \to -31.3 \pm 7.7$

Table 12: Joint ATR plus guard-side validation across the Gaussian sweep. Rows are split by the closed-loop recovery-band label. ATR is the radius term; SMPR and fixed-pool top-1 flip are guard-side criteria against task or action-candidate collapse and are not interpreted as standalone robustness metrics.

Task	ATR fragile → recovered	SMPR fragile → recovered	top-1 flip fragile → recovered
TwoRoom	1.00 → 0.12	0.36 → 0.98	0.79 → 0.07
PushT	0.72 → 0.10	0.85 → 1.00	0.38 → 0.06
Reacher	0.97 → 0.04	0.90 → 0.99	0.54 → 0.02
Cube	0.71 → 0.07	0.93 → 1.00	0.53 → 0.01
ALL	0.84 → 0.09	0.86 → 1.00	0.54 → 0.04

Table 13: Finite-difference Gaussian sensitivity analysis. The slope is the mean small-noise rollout radius squared divided by σ^2, summarized over the reported small-noise probes and training seeds. Lower endpoint/base ratios indicate reduced local composed encoder–predictor sensitivity. This is a local finite-difference proxy, not a global robustness guarantee.

Task	slope base → endpoint	endpoint/base	ATR base → endpoint
TwoRoom	4.16e+05 → 1.50e+03	0.004	1.51 → 0.11
PushT	1.87e+05 → 1.48e+03	0.008	3.58 → 0.25
Reacher	9.22e+04 → 1.64e+02	0.002	2.63 → 0.08
Cube	5.23e+04 → 4.40e+02	0.008	2.32 → 0.10

Table 14: Exact-JVP/Hutchinson local sensitivity decomposition. Values are endpoint/base ratios of Hutchinson trace estimates, using 16 sampled sequences and 8 Rademacher probes per checkpoint, then taking medians over training seeds. Columns report encoder trace per history-pixel dimension, rollout trace per latent-history dimension, composed encoder–rollout trace per history-pixel dimension, and the alignment coefficient $\mathrm{tr}(J_E^\top J_G^\top J_G J_E)/(\mathrm{tr}(J_E^\top J_E)\mathrm{tr}(J_G^\top J_G)/d_z)$. The analysis uses exact autograd JVPs with math/eager attention; it estimates local Frobenius traces, not a full Jacobian matrix or a closed-loop guarantee.

Task	encoder	rollout	composed	alignment
TwoRoom	0.004	1.256	0.003	0.644
PushT	0.017	0.995	0.010	0.633
Reacher	0.003	0.887	0.006	1.739
Cube	0.023	1.079	0.028	1.042

Table 15: Threshold sensitivity for recovered-vs-fragile diagnostic separation. Entries count tasks for which recovered rows have lower median normalized ATR and higher median SMPR than fragile rows under the specified behavioral-label definition.

Recovery fraction	clean tolerance	ATR pass tasks	SMPR pass tasks	rows
0.70	3	4/4	4/4	4
0.70	5	4/4	4/4	4
0.70	10	4/4	4/4	4
0.80	3	4/4	4/4	4
0.80	5	4/4	4/4	4
0.80	10	4/4	4/4	4
0.90	3	4/4	4/4	4
0.90	5	4/4	4/4	4
0.90	10	4/4	4/4	4

C Additional Gaussian Evaluation Tables

These tables report the full observation-only Gaussian evaluation columns available for the three-training-seed sweep behind Figure 1. Each training-seed value first averages evaluation seeds 42/43/44; table cells report mean ± population std across training seeds 3072/3073/3074.

Table 16: Full three-training-seed LeWM observation-only Gaussian sweep evaluation table. Columns report clean evaluation and observation-only Gaussian evaluation at $\sigma = 0.03/0.05/0.08$ with a clean goal image. Values are mean $\pm$ population std across training seeds 3072/3073/3074; each training-seed value first averages evaluation seeds 42/43/44 with 100 trajectories per evaluation seed.

Task	training $\sigma_{\max}$	clean eval	obs-only $\sigma = 0.03$ eval	obs-only $\sigma = 0.05$ eval	obs-only $\sigma = 0.08$ eval
TwoRoom	0.00	94.67 ± 0.72	92.56 ± 1.23	81.33 ± 6.77	68.78 ± 2.45
TwoRoom	0.01	94.89 ± 1.10	95.00 ± 0.82	94.78 ± 1.77	93.44 ± 2.08
TwoRoom	0.02	95.78 ± 1.10	95.89 ± 1.29	95.22 ± 1.29	95.44 ± 1.57
TwoRoom	0.03	95.11 ± 0.87	95.00 ± 0.82	95.22 ± 0.83	95.11 ± 1.34
TwoRoom	0.04	95.89 ± 0.42	96.44 ± 0.96	95.78 ± 0.31	95.89 ± 0.57
TwoRoom	0.05	96.11 ± 1.50	96.00 ± 1.91	96.00 ± 1.52	95.89 ± 1.55
TwoRoom	0.06	96.00 ± 1.70	95.56 ± 1.81	95.56 ± 1.40	95.78 ± 1.55
TwoRoom	0.07	96.44 ± 0.63	96.33 ± 0.54	96.11 ± 0.68	96.44 ± 0.31
TwoRoom	0.08	97.22 ± 0.83	97.22 ± 0.68	97.33 ± 0.72	97.11 ± 0.57
PushT	0.00	81.78 ± 4.89	53.56 ± 12.91	19.22 ± 9.12	7.22 ± 5.81
PushT	0.01	85.89 ± 1.50	82.22 ± 3.03	70.33 ± 6.33	37.89 ± 12.37
PushT	0.02	87.33 ± 0.72	85.11 ± 1.13	82.89 ± 2.69	62.67 ± 11.98
PushT	0.03	89.67 ± 0.00	89.67 ± 0.27	88.78 ± 2.08	82.44 ± 1.66
PushT	0.04	89.11 ± 0.83	88.22 ± 1.13	87.33 ± 1.19	86.56 ± 0.16
PushT	0.05	84.00 ± 3.03	84.22 ± 2.53	82.56 ± 3.13	81.11 ± 3.03
PushT	0.06	88.22 ± 1.34	88.89 ± 1.03	87.33 ± 0.47	86.78 ± 0.63
PushT	0.07	86.44 ± 0.87	86.00 ± 1.19	84.22 ± 0.42	84.00 ± 0.27
PushT	0.08	86.89 ± 1.59	86.56 ± 2.20	84.67 ± 2.88	85.78 ± 3.45
Reacher	0.00	59.22 ± 1.03	42.56 ± 2.99	33.56 ± 1.77	18.22 ± 0.42
Reacher	0.01	62.67 ± 1.19	60.67 ± 1.09	56.00 ± 0.72	44.56 ± 2.57
Reacher	0.02	77.00 ± 6.18	76.44 ± 8.17	77.22 ± 6.68	70.33 ± 9.46
Reacher	0.03	78.22 ± 1.66	79.33 ± 0.72	79.11 ± 1.10	76.78 ± 4.17
Reacher	0.04	81.78 ± 1.59	82.00 ± 1.52	81.11 ± 2.62	80.56 ± 1.03
Reacher	0.05	76.44 ± 4.57	77.33 ± 5.76	77.56 ± 4.01	77.78 ± 5.67
Reacher	0.06	82.22 ± 2.79	79.33 ± 2.42	81.67 ± 1.63	80.67 ± 1.19
Reacher	0.07	82.11 ± 2.20	82.44 ± 1.29	81.44 ± 1.85	83.33 ± 1.09
Reacher	0.08	81.33 ± 2.33	82.56 ± 2.08	82.00 ± 1.78	81.56 ± 0.87
Cube	0.00	65.22 ± 1.10	63.00 ± 1.19	46.67 ± 1.96	43.11 ± 3.27
Cube	0.01	66.78 ± 1.81	67.00 ± 2.13	67.56 ± 2.27	48.78 ± 0.31
Cube	0.02	63.67 ± 2.62	61.44 ± 3.45	63.33 ± 1.96	56.67 ± 3.07
Cube	0.03	64.56 ± 1.66	63.44 ± 3.03	65.33 ± 0.94	66.00 ± 1.70
Cube	0.04	68.11 ± 1.03	65.11 ± 1.50	65.44 ± 0.57	64.11 ± 0.83
Cube	0.05	62.67 ± 3.40	63.33 ± 2.42	60.67 ± 3.09	63.56 ± 2.53
Cube	0.06	66.00 ± 1.19	65.89 ± 1.03	64.67 ± 1.09	64.89 ± 1.85
Cube	0.07	64.67 ± 2.16	64.67 ± 1.89	65.44 ± 2.57	65.22 ± 1.55
Cube	0.08	61.67 ± 3.57	61.78 ± 4.16	61.89 ± 2.45	62.56 ± 1.55